ADITI AWASTHI

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Summary

PhD student researching compilation, resource estimation and error correction for fault-tolerant quantum computers (FTQC).

Education

PhD in Electrical and Computer Engineering Aug 2024 – May 2028 (Expected)
Advisor: Dr. Jonathan Baker; The University of Texas at Austin

MS in Computer Science Aug 2022 – May 2024
GPA: 3.82/4.00; The University of Texas at Austin

BE in Computer Science and Engineering Aug 2016 – Aug 2020
GPA: 9.83/10.00 (highest in CS 2020 cohort); B.M.S. College of Engineering, Bengaluru

Publications

Stalls and Spequlation: Pipelined Execution for Fault Tolerant Quantum Computation
DOI: 10.48550/arXiv.2606.19593
A. Awasthi, G.S. Ravi, and J.M. Baker — IEEE QCE '26

Price and Payoff: Non-Determinism in Fault Tolerant Quantum Computation
DOI: 10.48550/arXiv.2605.07983
A. Awasthi, S. Sethi, S. Khan, G.S. Ravi, and J.M. Baker — IEEE QCE '26

Staying Sober: A Shot-Free Predictor of Logical Error Rates for Concatenated Quantum Error Correcting Codes
DOI: 10.48550/arXiv.2610.06606
S. Sethi, **A. Awasthi**, M. Poster, J. Viszlai, and J.M. Baker — Under Review at QIP '27

Exploring Asymmetric QEC Code Concatenation
DOI: 10.48550/arXiv.2609.25251
S. Sethi, M. Poster, **A. Awasthi**, W. Yang, J. Viszlai, and J.M. Baker — QEC '26 (poster)

INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures
DOI: 10.48550/arXiv.2604.25094
S. Sethi, S. Khan, **A. Awasthi**, A. Anand, and J.M. Baker — IEEE QCE '26

Effectiveness of Connected Components Labelling Approach in Noise Reduction for Image De-fencing
DOI: 10.1007/978-981-16-3246-4_49
A. Awasthi, D. Bhat, M. Oak, and N. Kayarvizhy — ICCCT '21 (Springer)

LegalDB: Long DistilBERT for Legal Document Classification
DOI: 10.1109/ICAECT49130.2021.9392558
P. Bambroo and **A. Awasthi** — ICAECT '21

Professional Experience

Engineering Analyst, Synthetic Product Group, Goldman Sachs — Sep 2020 – Jun 2022
Built low-latency data pipelines for synthetic and swap reporting for quantitative trading clients

Engineering Intern, Synthetic Product Group, Goldman Sachs — Jan 2020 – Jun 2020
Engineering Intern, Ledgers Technology, Goldman Sachs — Jun 2019 – Aug 2019
Software Development Intern, Centre for Cyber Security, IIT Kanpur — Jun 2018 – Jul 2018

Research Experience

Dr. Jonathan Baker's Lab, UT Austin — Aug 2024 – Present
Resource estimation, speculative and pipelined execution, and decoding for FTQC
Collaborating with Dr. Poulami Das on resource allocation for distributed quantum computing

Dr. David Soloveichik's Lab, UT Austin — Jun 2023 – May 2024
Unconventional computing, molecular programming

Dr. Kayarvizhy N's Lab, B.M.S. College of Engineering — Aug 2019 – Aug 2020
Image de-fencing using frequency-domain analysis and connected component labelling

Awards

Charles M. Simmons Endowed Presidential Fellowship in Engineering — Aug 2026 – May 2027
IEEE Quantum Week Student Travel Grant — Sep 2026
BMSCE Alumni Network Merit Scholarship (6-time recipient) — Aug 2016 – Aug 2020

Teaching

- Grad TA for ECE 412 - Software Design and Implementation I — Fall 2026
  Delivered a guest lecture on 'Choosing Algorithms based on Input Structure'
- Grad TA for ECE 381V/CS 395T - Unconventional Computation — Spring 2024, 2025, 2026
- Grad TA for ECE 461L - Software Engineering and Design Lab — Fall 2025
- Grad TA for CS 388G - Algorithms: Techniques and Theory — Spring 2023
- Instructor for IEEE Special Interest Group - Programming with Python — Fall 2017

Extracurriculars

- Vice-Chairperson, BMSCE IEEE Student Branch — Aug 2018 – Jul 2019
- Department Coordinator, PhaseShift Technical Fest, BMSCE — Aug 2018 – Sep 2018

Technical Skills

Quantum Computing: IBM Qiskit, Stim, Sinter
Programming: Python, Java, C++, R, SQL, Linux Shell
Tools & Frameworks: MongoDB, Hadoop, Apache Spark, React.js, Git